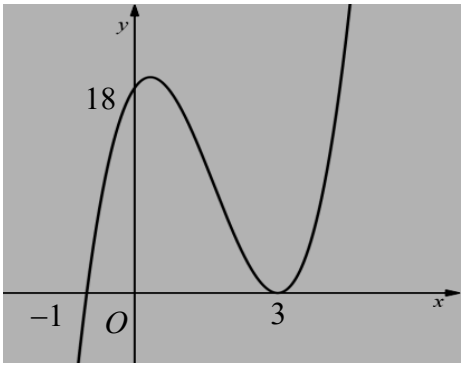
1.	$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4}\,dx = 12 \times \frac{x^4}{4} + \frac{1}{6} \times 2x^{\frac{1}{2}} - \frac{3}{2} \times \frac{x^{-3}}{-3}$	M1
	$= 3x^4 + \frac{1}{3}x^{\frac{1}{2}} + \frac{1}{2}x^{-3} + c$	A1A1A1A1
		(5)
		(5 marks)

2.	$y = 3x^5 + 4x^3 - x + 5 \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}x} =\right) 15x^4 + 12x^2 - 1$	M1 A1
	$15x^4 + 12x^2 - 1 = 2 \Rightarrow 15x^4 + 12x^2 - 3 = 0$	dM1
	$\Rightarrow 3(5x^2 - 1)(x^2 + 1) = 0$ o.e	ddM1
	$\Rightarrow x = \pm\frac{1}{\sqrt{5}}$ o.e.	A1
		(5)
		(5 marks)

3.(i)	$\frac{3}{x} > 4 \Rightarrow 3x > 4x^2 \Rightarrow x(4x-3) < 0 \Rightarrow 0, \frac{3}{4}$	B1
	$0 < x < \frac{3}{4}$	M1 A1
		(3)
(ii)	$y - 0 = 3(x+5)$	B1
	E.g. $y < 2x^2 - 50,\ y > 3x + "15"$	M1
	E.g. $y < 2x^2 - 50,\ y > 3x + 15,\ x < -5$	A1
		(3)
		(6 marks)

4.(a)(i)	$(90,-1)$	B1 B1
(ii)	225	B1
		(3)
(b)	One of $-1 < p < 0$, $p = 1$	M1
	Both $-1 < p < 0$, $p = 1$	A1
		(2)
		(5 marks)

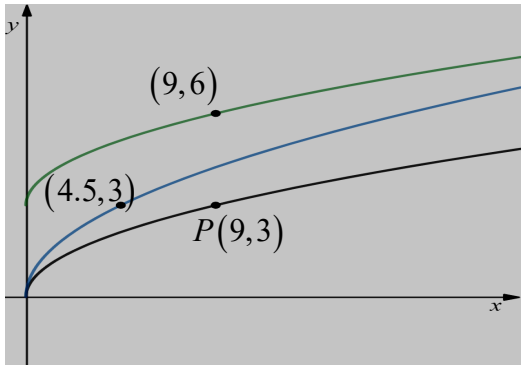
5(a)	$3y-2x=30 \Rightarrow m=\frac{2}{3}$	B1
	$y=-\frac{3}{2}(x-24),\ y=-\frac{3}{2}x+36,\ 2y+3x=72,\ \frac{y-0}{x-24}=-\frac{3}{2}$	M1 A1ft
	Full method to find one co-ordinate of P E.g. Solves $\frac{2}{3}x+10=-\frac{3}{2}(x-24)$	M1
	Coordinates of P $(12,18)$	A1
		(5)
(b)	$y=0,\ 3y-2x=30 \Rightarrow x=...$	M1 (B1 on EPEN)
	Area $BPA=\frac{1}{2}\times(24+"15")\times"18"=351$	dM1 A1 cso
		(3)
		(8 marks)

6.(a)		Positive cubic shape anywhere with 1 maximum and 1 minimum	M1
		Positive cubic shape that at least reaches the x-axis at $x = -1$ and with a minimum on the x-axis at $x = 3$	A1
		y intercept at 18. Must correspond with their sketch	B1
	For the intercepts allow as numbers as above or allow as coordinates e.g. (18, 0), (0, −1), (0, 3) as long as they are marked in the correct place.		
			(3)
(b)	E.g. $(2x+2)(x^2-6x+9) = \ldots$		M1
	$= 2x^3 - 10x^2 + 6x + 18$		A1 A1
			(3)
(c)	$\left(f'(x) =\right) 6x^2 - 20x + 6$		B1ft
	$f'\left(\frac{1}{3}\right) = 6\left(\frac{1}{3}\right)^2 - 20\left(\frac{1}{3}\right) + 6$		M1
	$f'\left(\frac{1}{3}\right) = 0$		A1
	$y = \frac{512}{27}$		A1
			(4)
			(10 marks)

7.(a)	$OB^2 = 0.6^2 + 1.4^2 - 2\times 0.6\times 1.4\cos 2 \Rightarrow OB^2 = ...$ or $OB = ...$	M1
	$OB = 1.738$	A1
	$\dfrac{\sin AOB}{1.4} = \dfrac{\sin 2}{"1.738"} \Rightarrow AOB = 0.822$ or e.g. $\dfrac{\sin ABO}{0.6} = \dfrac{\sin 2}{"1.738"} \Rightarrow ABO = 0.319... \Rightarrow AOB = \pi - 2 - 0.319...$	dM1
	$\theta = 2\times AOB = 2\times 0.822 = 1.64*$	A1*
		(4)
(b)	Attempts $0.6\times\alpha$ with $\alpha = 2\pi - 1.64$ or $\alpha = \pi - 1.64$ or Attempts $2\times\pi\times 0.6 - 0.6\times 1.64$	M1
	$0.6\times(2\pi - 1.64) + 2.8 = 5.6\,\text{m}$	A1
		(2)
(c)	Attempts $\dfrac{1}{2}\times 0.6^2\times\alpha$ with $\alpha = 2\pi - 1.64$ or $\alpha = \pi - 1.64$ or Attempts $\pi\times 0.6^2 - \dfrac{1}{2}\times 0.6^2\times 1.64$	M1
	Attempts $0.6\times 1.4\sin 2$	M1
	Full method $\dfrac{1}{2}\times 0.6^2\times(2\pi - 1.64) + 0.6\times 1.4\sin 2 = 1.6\,\text{m}^2$	ddM1 A1
		(4)
		(10 marks)

(a)	$OB^2 = 0.6^2 + 1.4^2 - 2\times 0.6\times 1.4\cos 114.59... \Rightarrow OB^2 = ...$ or $OB = ...$	M1
	$OB = 1.738$	A1
	$\frac{\sin AOB}{1.4} = \frac{\sin 114.59...}{"1.738"} \Rightarrow AOB = 0.822$ or e.g. $\frac{\sin ABO}{0.6} = \frac{\sin 114.59...}{"1.738"} \Rightarrow ABO = 18.3 \Rightarrow AOB = 180 - 114.59... - 18.3$	M1
	$\theta = 2\times AOB = 2\times 47.1 = 94.2° = 1.64$ *	A1*
		(4)
(b)	Attempts $\frac{\alpha}{360}\times 2\times\pi\times 0.6$ with $\alpha = 360 - \text{awrt } 94°$ or Attempts $2\times\pi\times 0.6 - \frac{\text{awrt } 94°}{360}\times 2\times\pi\times 0.6$	M1
	$\frac{360 - \text{awrt } 94}{360}\times 2\times\pi\times 0.6 + 2.8 = 5.6\,\text{m}$	A1
		(2)
(c)	Attempts $\frac{\alpha}{360}\times\pi\times 0.6^2$ with $\alpha = 360 - \text{awrt } 94°$ or Attempts $\pi\times 0.6^2 - \frac{\text{awrt } 94°}{360}\times\pi\times 0.6^2$	M1
	Attempts e.g. $0.6\times 1.4\sin 114.59...$	M1
	Full method e.g. $\frac{360 - \text{awrt } 94°}{360}\times\pi\times 0.6^2 + 0.6\times 1.4\sin 114.59... = 1.6\,\text{m}^2$	dM1 A1
		(4)
		(10 marks)

8.(a)(i)	$4+12x-3x^2=a\pm3\left(x+c\right)^2$ or $a+b\left(x\pm2\right)^2$	M1
	Two of $16-3\left(x-2\right)^2$ or two of $a=16, b=-3, c=-2$	A1
	$16-3\left(x-2\right)^2$	A1
(ii)	Coordinates $M=\left(2,16\right)$	B1ft B1ft
		(5)
(b)	States or implies that l_2 has equation $y="8"x+k$	M1
	Sets $4+12x-3x^2="8x"+k$ and proceeds to 3TQ	dM1
	Correct 3TQ $3x^2-4x+k-4=0$	A1
	Attempts to use $b^2-4ac=0$ to find k	ddM1
	$k=\frac{16}{3}\Rightarrow y=8x+\frac{16}{3}$	A1
		(5)
		(10 marks)

9(a)		
	One correct sketch drawn and labelled correctly	M1
	One correct sketch drawn and labelled and with correct point	A1
	Completely correct sketches with both points	A1
		(3)
(b)	Sets $\sqrt{x} + 3 = \sqrt{2x}$	B1
	$3 = \left(\sqrt{2} - 1\right)\sqrt{x}$	M1
	$\sqrt{x} = \dfrac{3}{\left(\sqrt{2}-1\right)} \times \dfrac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}+1\right)} = 3\left(\sqrt{2}+1\right)$ *	A1*
		(3)
(c)	$\sqrt{x} = 3\left(\sqrt{2}+1\right) \Rightarrow x = 9\left(\sqrt{2}+1\right)^2 = \ldots$	M1
	$\Rightarrow x = 9\left(3 + 2\sqrt{2}\right),\ y = 3\sqrt{2} + 6$	A1, B1
		(3)
		(9 marks)

10(a)	$f'(x) = ax - 12x^{\frac{1}{3}} \Rightarrow f''(x) = a - 4x^{-\frac{2}{3}}$	B1
	Sets $f''(27) = 0 \Rightarrow 0 = a - 4 \times \frac{1}{9} \qquad \Rightarrow a = \frac{4}{9}$	M1 A1
		(3)
(b)	$f'(x) = ax - 12x^{\frac{1}{3}} \Rightarrow \left(f(x) = \right)\frac{1}{2}ax^{2} - 9x^{\frac{4}{3}} + c$	M1 A1ft
	Substitutes $x = 1, f(x) = -8 \Rightarrow c = ...$	dM1
	$\left(f(x) = \right)\frac{2}{9}x^{2} - 9x^{\frac{4}{3}} + \frac{7}{9}$	A1
		(4)
		(7 marks)